

MADRUSHKAT, TajikistanWMO: **387250**Lat: **39.433N**Lon: **69.667E**Elev: **2234**StdP: **77.21**Time Zone: **5.00 (E05)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-10.7	-25.6	0.5	-8.9	-23.5	0.6	-7.5	7.4	0.9	6.7	0.3	1.0	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	29.5	14.1	28.0	13.5	26.4	12.8	15.1	27.2	14.2	25.8	13.4	24.4	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.2	10.9	17.0	10.1	10.1	16.5	9.1	9.4	16.0	50.7	26.9	47.6	26.0	45.1	24.6	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.1	5.3	-15.1	32.4	2.1	1.3	-16.6	33.3	-17.8	34.1	-19.0	34.8	-20.4	35.7		
(5)			-16.3	17.6	1.8	1.1	-17.6	18.5	-18.6	19.1	-19.6	19.8	-20.9	20.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-2.3	-1.3	4.3	9.2	13.0	16.0	19.2	18.7	15.2	9.2	3.2	-1.0
	DBStd	8.23	3.19	4.02	3.81	3.46	2.69	2.73	2.76	2.66	2.88	3.43	3.40	3.62
	HDD10.0	1539	380	316	179	56	6	1	0	0	3	56	203	341
	HDD18.3	3627	638	549	435	275	166	79	23	28	98	283	453	599
	CDD10.0	1056	0	0	2	31	99	180	286	268	157	31	1	0
	CDD18.3	102	0	0	0	0	1	9	51	38	3	0	0	0
	CDH23.3	1334	0	0	0	3	22	133	596	482	96	3	0	0
	CDH26.7	294	0	0	0	0	1	16	161	108	8	0	0	0
Wind	WSAvg	1.7	2.0	2.2	2.3	1.9	1.7	1.6	1.3	1.2	1.3	1.4	1.5	1.6
Precipitation	PrecAvg	351	24	31	46	55	55	29	20	10	9	18	26	28
	PrecMax	548	43	71	85	122	139	67	74	31	43	49	81	62
	PrecMin	113	3	11	4	9	6	9	3	2	2	4	4	2
	PrecStd	124	10	14	20	30	33	16	18	9	9	10	18	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	12.7	17.1	23.1	25.8	28.4	31.7	30.8	27.9	23.0	14.9	10.0
		MCWB	-0.2	2.7	5.6	8.9	10.4	13.1	15.4	14.5	12.0	9.6	4.7	1.5
	2%	DB	5.2	8.0	14.2	20.1	23.3	26.4	30.0	29.3	25.9	20.1	12.1	7.6
		MCWB	-1.3	1.3	4.2	7.7	9.8	12.4	14.5	13.9	11.3	8.2	3.5	0.4
	5%	DB	4.0	5.9	12.3	17.9	21.5	24.7	28.5	27.8	24.2	17.8	10.4	5.9
		MCWB	-1.9	0.3	3.3	6.6	9.0	11.7	14.1	13.3	10.7	7.1	2.5	-0.1
	10%	DB	2.8	4.2	10.6	15.9	19.7	23.0	26.8	26.2	22.5	15.9	8.7	4.4
		MCWB	-2.3	-0.5	2.7	5.8	8.3	11.1	13.5	12.7	10.2	6.2	1.5	-0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	5.0	6.6	9.6	11.7	14.3	16.8	15.7	13.1	10.2	6.1	3.7
		MCDB	4.1	9.0	12.7	19.0	20.5	24.4	27.6	28.0	24.6	21.3	12.8	5.8
	2%	WB	-0.1	3.1	5.3	8.6	10.7	13.2	15.7	14.6	11.9	8.8	4.5	2.3
		MCDB	3.2	5.9	11.2	16.8	19.2	23.9	28.0	26.8	24.4	18.5	9.6	5.5
	5%	WB	-0.9	1.6	4.4	7.7	10.1	12.4	14.7	13.8	11.2	7.8	3.6	0.6
		MCDB	2.4	4.6	9.9	14.7	17.8	22.4	26.6	25.7	22.8	15.7	8.5	5.1
	10%	WB	-1.7	0.3	3.6	6.8	9.4	11.6	13.9	13.1	10.4	6.9	2.5	-0.5
		MCDB	1.8	3.5	8.7	13.5	16.5	20.9	25.1	24.3	21.2	13.8	7.1	3.8
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.6	9.4	10.8	11.7	13.0	14.2	14.6	13.9	11.8	9.7	8.6
		MCDBR	9.5	10.0	11.7	13.4	14.7	15.4	16.4	16.5	16.3	14.4	11.7	10.2
		MCWBR	6.0	6.0	5.6	5.9	5.4	5.3	6.0	5.8	6.0	6.2	6.3	6.4
	5% WB	MCDBR	7.6	8.7	9.8	11.7	12.3	13.7	15.4	15.2	15.2	13.0	9.8	8.7
		MCWBR	5.1	5.8	5.1	5.4	5.0	5.2	6.2	5.6	6.1	5.8	5.4	5.8
Clear-Sky Solar Irradiance	taub	0.282	0.320	0.391	0.419	0.436	0.447	0.446	0.408	0.376	0.385	0.334	0.278	
	taud	2.213	2.097	1.971	1.989	2.013	2.000	2.015	2.133	2.180	2.067	2.145	2.252	
	Ebn at Noon	880	886	853	855	847	835	831	855	861	798	798	856	
	Edh at Noon	99	130	165	173	173	175	171	148	132	133	106	88	
All-Sky Solar Radiation	RadAvg	2.19	2.96	3.97	4.84	5.80	6.90	7.18	6.82	5.67	3.68	2.22	1.83	
	RadStd	0.17	0.38	0.48	0.71	0.52	0.41	0.36	0.42	0.27	0.45	0.34	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	+0.49	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page